SOP: Configure VNet-to-VNet VPN Connectivity in Microsoft Azure

Objective

To establish secure connectivity between two Azure Virtual Networks using Azure Virtual Network Gateways and validate communication between virtual machines using private IP addresses.

Step 1: Create Virtual Networks

Azure Portal → Virtual Networks
Created two Virtual Networks in the same Resource Group.

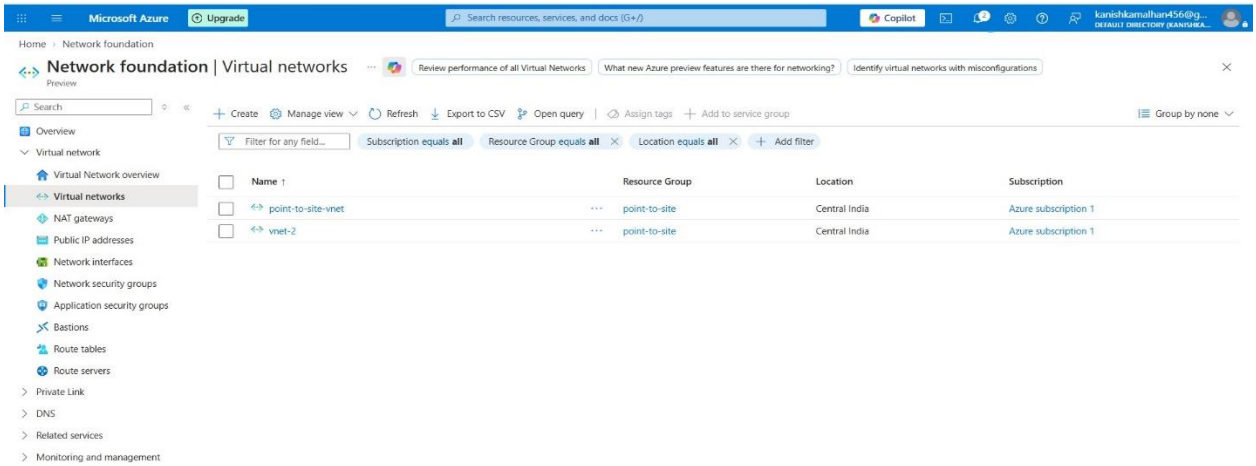
Configuration

VNet-1 : point-to-site-vnet
VNet-2 : vnet-2

Resource Group : point-to-site
Region : Central India

Observation

Both virtual networks were successfully created with separate address spaces to support VNet-to-VNet connectivity.



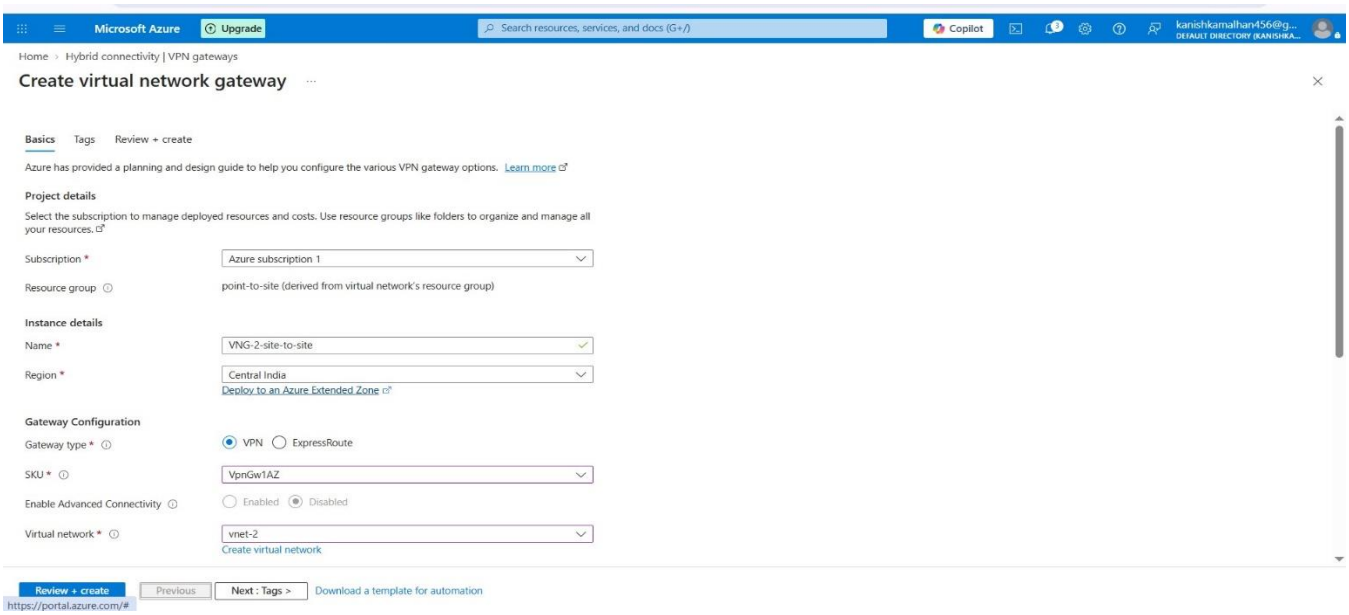
Step 2: Create Virtual Network Gateway

Azure Portal → VPN Gateways → Create

Configured a Virtual Network Gateway for the second virtual network.

Observation

The Virtual Network Gateway was configured and associated with vnet-2.

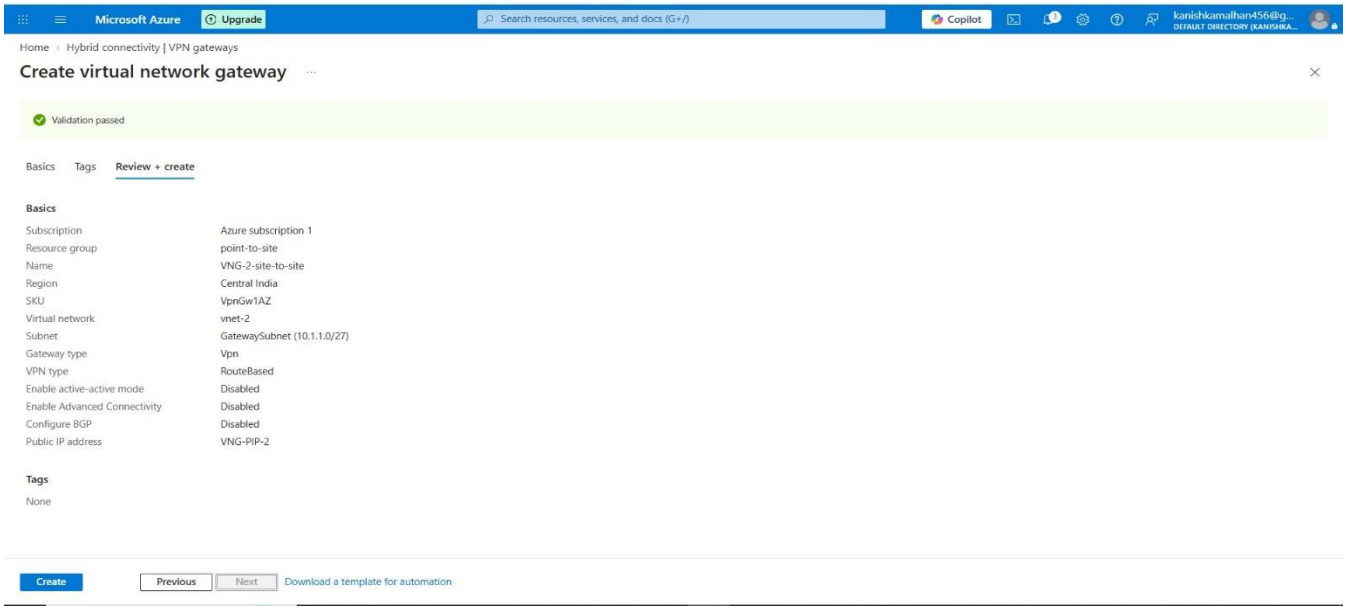


Step 3: Review and Deploy Gateway

Reviewed the gateway configuration and validated all deployment settings before creating the resource.

Observation

Validation completed successfully and deployment was initiated.



Step 4: Verify Virtual Network Gateways

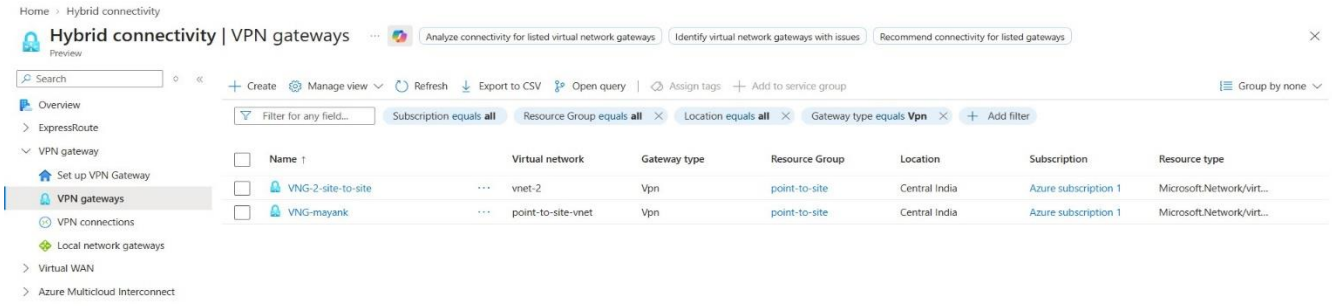
After deployment, verified that both Virtual Network Gateways were available.

Gateways

VNG-mayank
VNG-2-site-to-site

Observation

Both gateways were successfully deployed and ready for VPN configuration.



Step 5: Create VNet-to-VNet VPN Connection

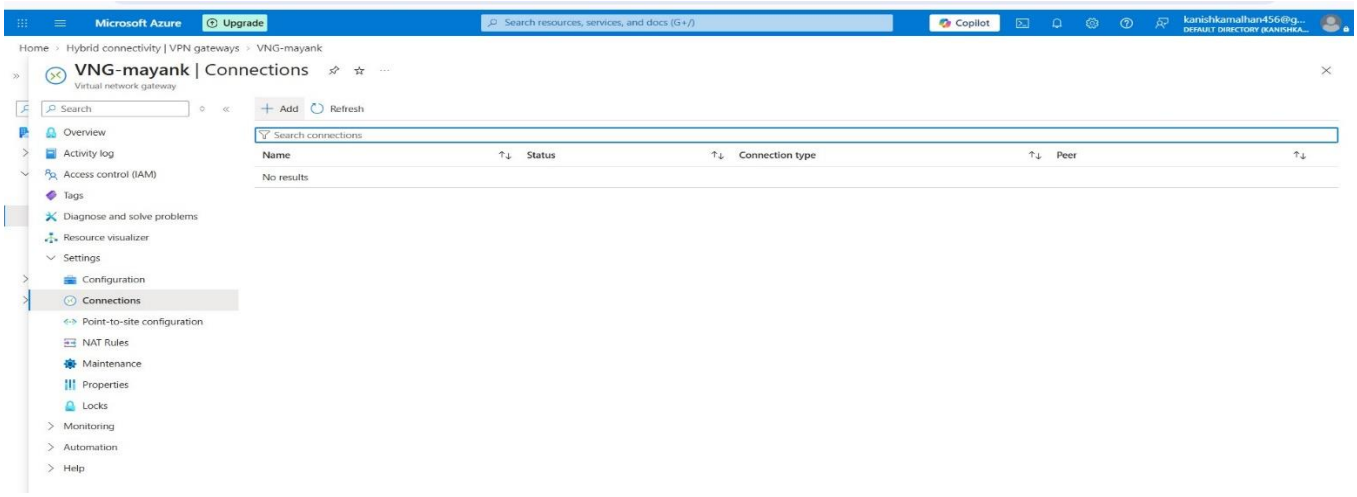
Navigate to:

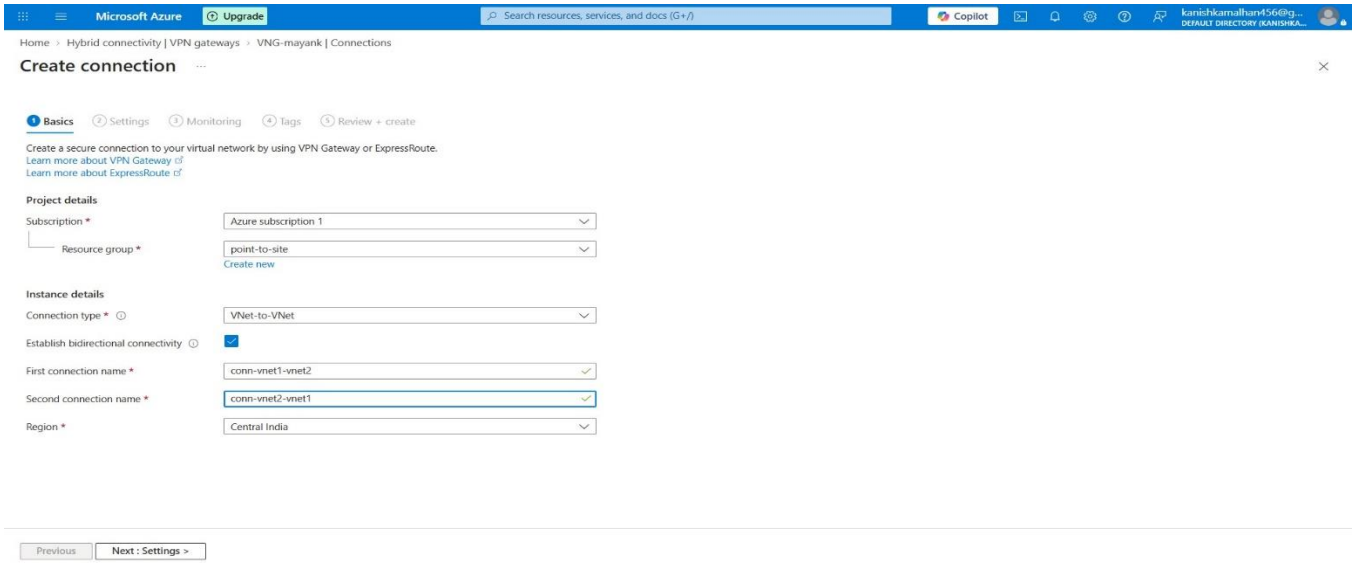
VNG-mayank
→ Connections
→ Add

Created a new VPN connection between both virtual network gateways.

Observation

A bidirectional VNet-to-VNet VPN connection was configured between both gateways.



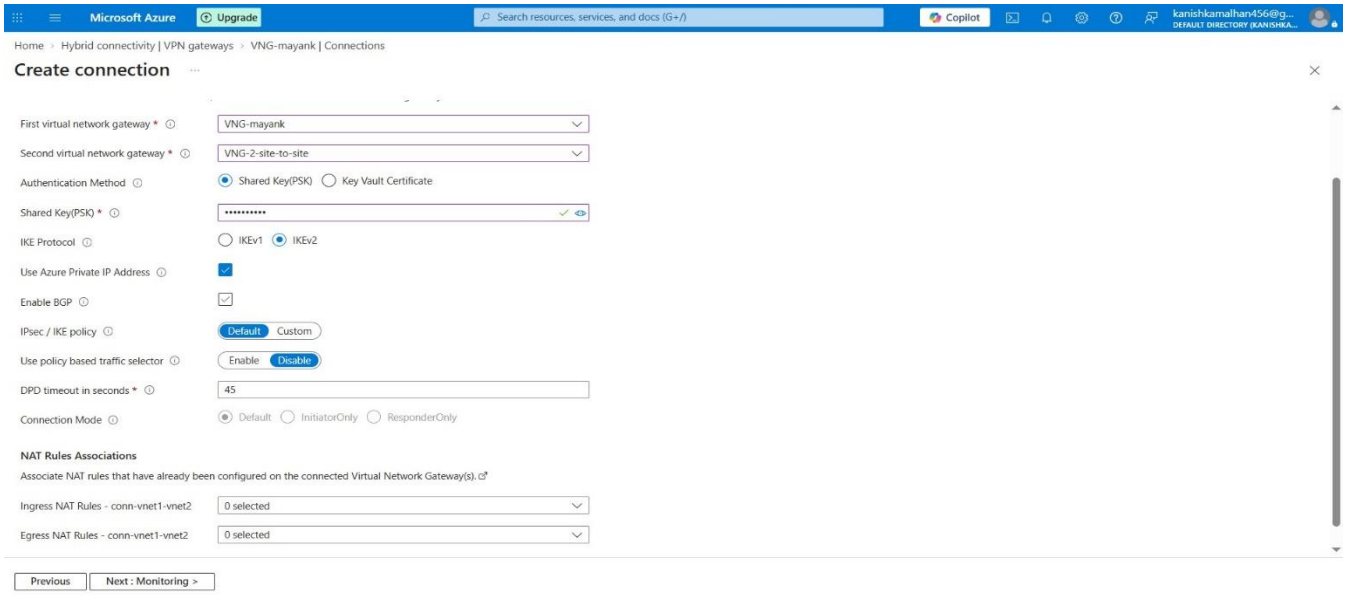


Step 6: Configure VPN Connection Settings

Configured the VPN connection settings and selected both gateways for communication.

Observation

VPN authentication and tunnel settings were configured successfully.



Step 7: Verify VPN Connection Status

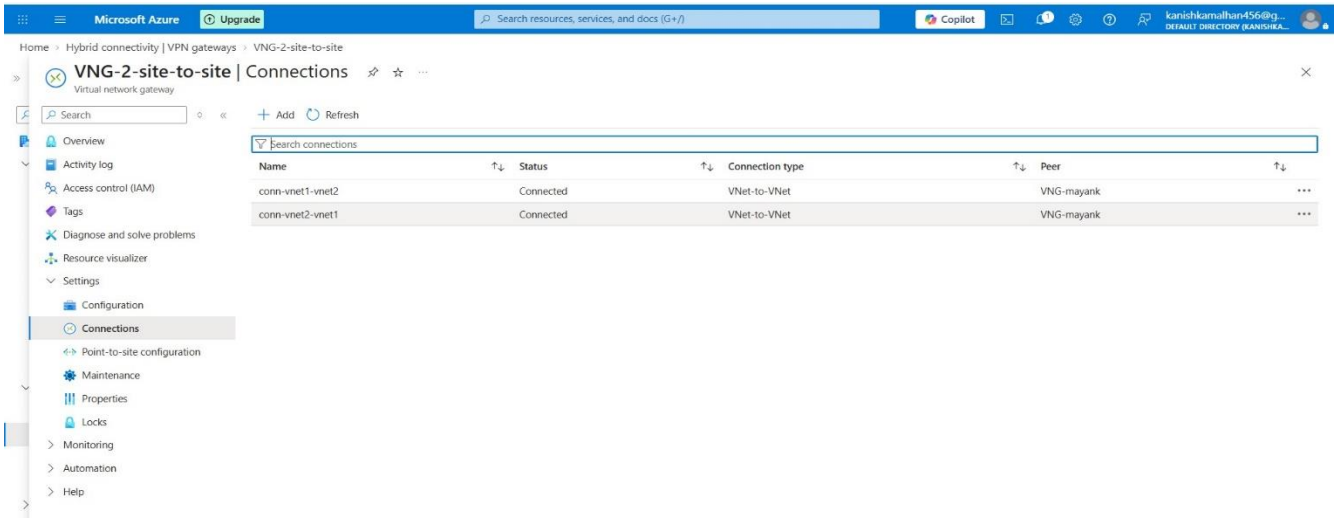
Verified the status of the VPN tunnels after deployment.

Result

Plain Text
conn-vnet1-vnet2 : Connected
conn-vnet2-vnet1 : Connected

Observation

Both VPN tunnels displayed the **Connected** status, confirming successful VNet-to-VNet connectivity.

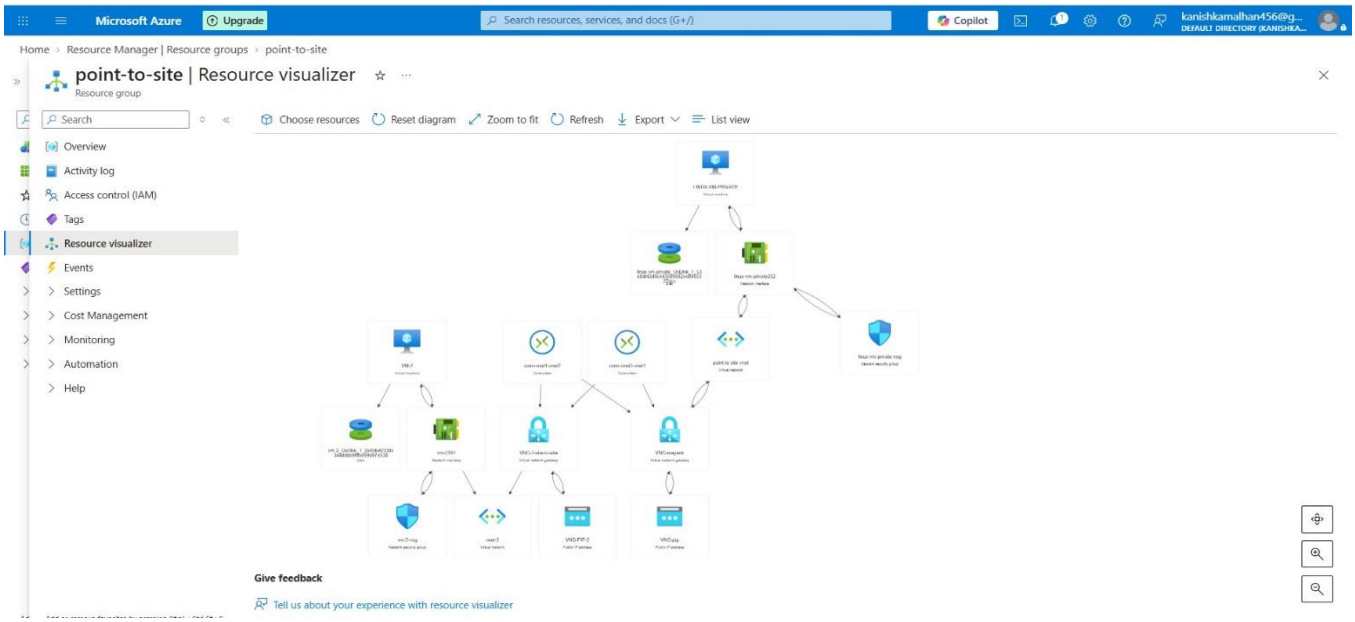


Step 8: Validate Deployed Resources

Opened the Resource Visualizer to verify connectivity between deployed resources.

Observation

The Resource Visualizer confirmed successful deployment and integration of both VNets, gateways, VPN connections, and virtual machines.



Step 9: Validate SSH Connectivity

Connected from VM-1 to VM-2 using the private IP address through the VNet-to-VNet VPN tunnel.

Command

ssh azureuser2@10.1.0.4

Result

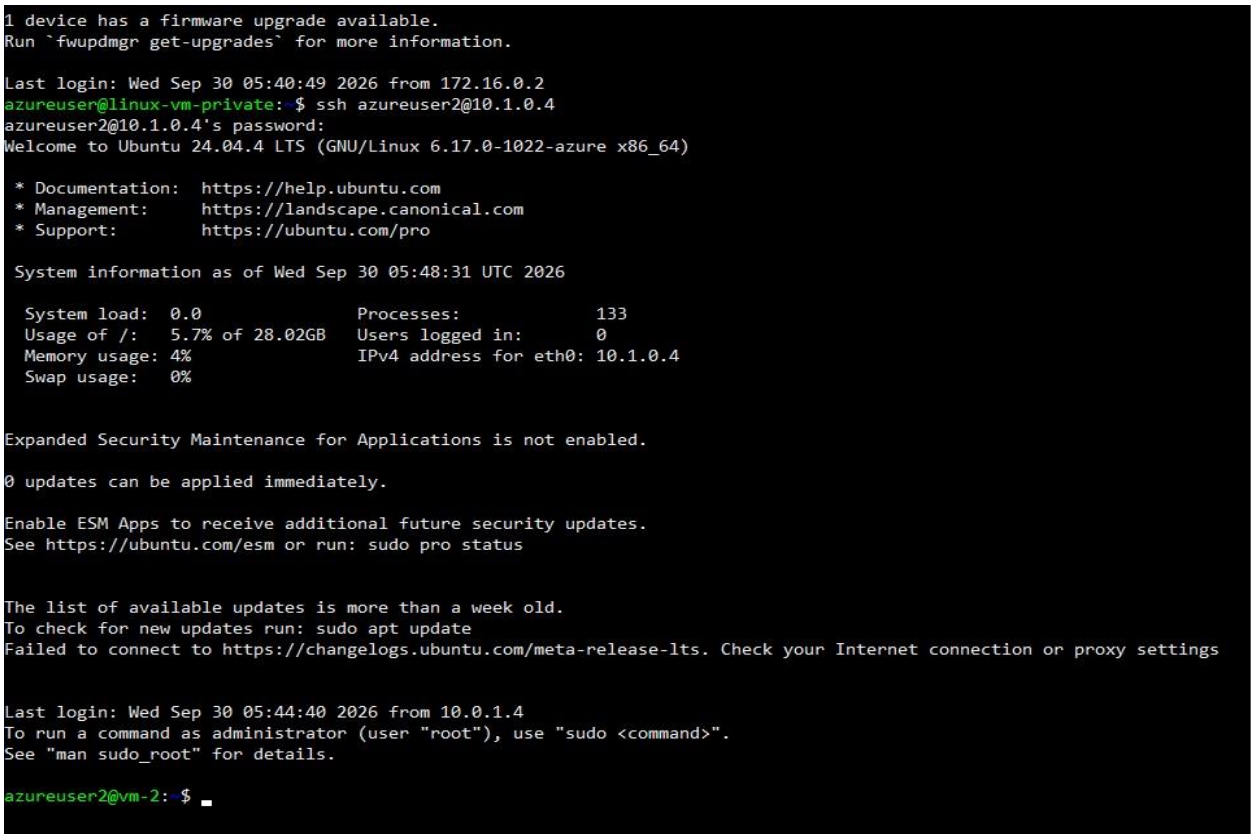
SSH Connection Successful

Successfully connected from:
linux-vm-private (VM-1)

To:
vm-2 (10.1.0.4)

Observation

Successful SSH login confirmed end-to-end communication between virtual machines deployed in different virtual networks through the VPN tunnel.



Final Observation

Successfully established a VNet-to-VNet VPN connection between **point-to-site-vnet** and **vnet-2** using Azure Virtual Network Gateways. The VPN connection status was verified as **Connected** on both gateways. Connectivity validation was successfully performed by establishing an SSH connection from **linux-vm-private (VM-1)** to **vm-2 (10.1.0.4)** using the private IP address through the VPN tunnel.

Result

Created two Virtual Networks successfully
Configured and deployed Virtual Network Gateways successfully
Established VNet-to-VNet VPN connectivity successfully
Verified VPN tunnel status as Connected
Validated deployed resources using Resource Visualizer
Successfully connected VM-1 to VM-2 using SSH over private IP
Successfully verified secure communication between both VNets